

COMPUTER SCIENCE TRIPOS Part IB – 2020 – Paper 6

10 Logic and Proof (lp15)

propositional
logic

- (a) From n distinct propositional letters, each of which may be negated or not, 2^n distinct clauses can be created. Present a satisfying interpretation of this set of 2^n clauses or demonstrate that none exists. [3 marks]

Answer: None exists, as is strongly hinted by smaller instances. (For $n = 0$ we have only the empty clause.) There are 2^n possible propositional interpretations, each of which is excluded by one of the 2^n clauses. E.g. for $n = 2$, there are four clauses; the clause $\{P, Q\}$ excludes the interpretation that sets P and Q both to false, etc. One can also identify the falsified clause by negating all the literals assigned by a given interpretation.

clausal proof
methods

- (b) Sketch the operation of the DPLL algorithm when provided with the set of clauses described above, including an estimate of its time complexity as a function of n . [4 marks]

Answer: There can be no pure literals, as each variable appears with both signs. Except with $n = 1$, there are no unit clauses. Therefore a case split is necessary. If the propositional letters are $P_1, \dots, P_n$, we can arbitrarily assume that P_n will be chosen (the problem is symmetric in all variables). In each of the two cases, P_n will be eliminated from the set of clauses, leaving a similar set of clauses with $n - 1$ variables. Therefore 2^n case splits can be expected. (Note that the total time complexity is trivially at least $O(2^n)$ due to the sheer size of the problem. The real intent of the question concerned an estimate of the number of case splits.)

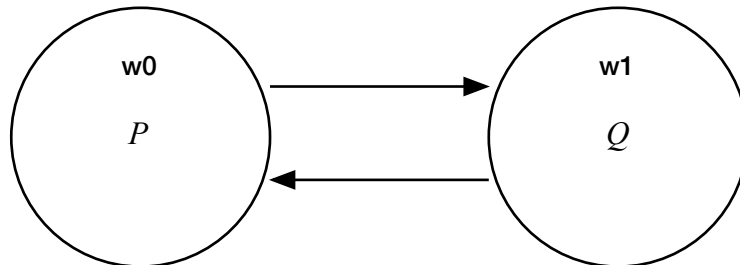
First-Order Logic,
Sequent
Calculus/
Tableaux
Methods, Modal
Logic

- (c) For each of the following formulas, present either a proof in the sequent calculus, or a falsifying interpretation. The modal logic is S4.

(i) $\Box(P \vee Q) \rightarrow (\Box \Diamond \neg P \rightarrow \Diamond \Box Q)$ [5 marks]

(ii) $\exists x P(f(x)) \wedge \forall x [P(x) \rightarrow Q(g(x))] \rightarrow \exists y Q(y)$ [4 marks]

(iii) $\exists x (P(x) \rightarrow Q(x)) \rightarrow [\exists x P(x) \rightarrow \exists x Q(x)]$ [4 marks]



Answer:

- (i) Consider two interconnected worlds: w_0 , where only the propositional letter P is true, and w_1 , where only the propositional letter Q is true (see diagram). Then $P \vee Q$ holds in both worlds, and therefore so does $\Box(P \vee Q)$; similarly $\Diamond \neg P$ and $\Box \Diamond \neg P$ hold in both worlds. But in no world do we have $\Box Q$, so also not $\Diamond \Box Q$. An infinite chain of worlds is also acceptable.

- (ii) Working backwards, the steps of the sequent proof are as follows:

$$\begin{aligned}
 & \exists x P(f(x)) \wedge \forall x [P(x) \rightarrow Q(g(x))] \rightarrow \exists y Q(y) && \text{original formula} \\
 & P(f(x)), \forall x (P(x) \rightarrow Q(g(x))) \Rightarrow \exists y Q(y) && (\wedge l), (\exists l), (\rightarrow r) \\
 & P(f(x)), P(f(x)) \rightarrow Q(g(f(x))) \Rightarrow \exists y Q(y) && (\forall l) \\
 & P(f(x)), P(f(x)) \rightarrow Q(g(f(x))) \Rightarrow Q(g(f(x))) && (\exists r)
 \end{aligned}$$

It's essential to eliminate the quantifiers in the correct order. The continuation is $(\rightarrow l)$ with two basic sequents.

- (iii) This formula is not valid. Note that if Q is interpreted by the constantly false formula then the formula simplifies to $\exists x \neg P(x) \rightarrow \neg \exists x P(x)$ and it can be falsified in a model over the integers, interpreting $P(x)$ by $x \neq 0$.
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